When Verifiers Backfire: A Cost–Correctness Theory of Generation–Verification–Repair Loops for LLMs

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October 2026

Abstract

Test-time verification is rapidly becoming a standard component of LLM agent systems: a generator produces an artifact, a verifier accepts or rejects it, and rejections trigger repair. Existing theory (Dorner et al., ICLR 2026) characterizes how verifier imperfection shapes pure resampling methods such as rejection sampling and Best-of-N. We extend this line of inquiry along three axes that pure-resampling theory does not cover: (i) a *miss-asymmetry theorem* giving the closed-form exchange rate $R = a/(1-b)$ between the two verifier error types when retries redraw at the generator's rate—misses hurt final precision $a/(1-b)$ times as much per unit as false rejections, whose cost lands almost entirely in extra attempts; (ii) a *repair-degradation theorem* showing the exchange rate is modulated by the retry success rate q: when q falls below a computable threshold q^*, false rejections become the dominant harm and the asymmetry *flips*; and (iii) an *attribution analysis* decomposing the terminal-error underprediction of pooled per-attempt operating points into degraded retries (the dominant channel) and error clustering (a residual). We validate the theory on HumanEval (164 problems, six arm configurations), pairing every looped arm's terminal outcome against its own first attempt. Oracle (unit-test) verification never loses a correct solution and rescues 14/10 wrong ones (McNemar $p < 10^{-3}$); LLM judges with pooled operating points $(a, b) \approx (0.78\text{–}0.86, 0.74\text{–}0.86)$ do the reverse—net -12 correct solutions for the isolated self-judge ($p = 0.004$, robust to counting only delivered outcomes), -8 for the cross-capability judge ($p = 0.039$, convention-dependent), and a non-significant -4 for the context-sharing judge—and the flip is predicted from measured parameters alone ($q \approx 0.3$ vs. $q^* \approx 0.6\text{–}0.75$). A budget sweep sharpens the practical rule: every judge arm's single-shot delivered stream is purer than no verification, and the first retry already erases the gain for the self-judge. The $\approx 3.4\times$ pooled-rate underprediction of terminal leak (task-level bootstrap 95% CI $[2.1, 4.7]$) is fully explained by degraded retries: pooled rates mask the per-round trajectory, and no residual requires error clustering. A matched-pool ablation *suggests* that actionable feedback raises repair success (0.33 vs. 0.19; small sample), and we report an honest negative result on verifier-context self-verification collapse. Together, the results turn "add a verifier" from an article of faith into a quantitative design decision.

1 Introduction

Production LLM agent systems increasingly wrap generation in a verification loop: an executor produces an artifact, a verifier judges it, rejections trigger repair, and the loop terminates when the verifier accepts or the budget runs out. This generation–verification–repair (GVR) pattern underlies executor/ verifier pipelines, LLM-as-judge filters, and test-driven agent loops. The engineering

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question it raises is sharp and often ignored: *when does adding a verifier actually improve the final answer, and when does it make things worse?*

Two literatures border this question without answering it. The self-correction literature documents empirically that intrinsic self-correction frequently fails or degrades performance (Stechly et al., 2025; Kamoi et al., 2024; Huang et al., 2024), while sound external verification helps. The test-time-scaling theory of Dorner et al. (2026) proves that instance-level accuracy of rejection sampling and Best-of-N is characterized by the geometry of the verifier's ROC curve—but their analysis covers *pure resampling*: rejection carries no feedback, retries are independent redraws, and no guidance is given on where along the ROC curve a verifier should operate.

This paper contributes a compact theory of the GVR loop and a 164×6 empirical study that stresses it. Our contributions:

1. **Miss-asymmetry theorem (§3.2).** For a GVR loop with generator success p and verifier operating point (a, b), the marginal damage of the miss rate $1 - b$ to final precision exceeds that of the false-rejection rate $1 - a$ by the factor $R = a/(1 - b)$. When retries redraw at the generator's rate, false rejections are nearly free for correctness and are paid for only in attempts ($R \approx 19$ at $(0.95, 0.95)$): push catch rate aggressively. The advice is conditional—§3.5 derives the threshold q^* below which it inverts.

2. **Repair-degradation theorem (§3.5).** The exchange rate above assumes retries redraw at the original success rate. Real repair attempts are degraded—conditioned on rejection, retries draw from a much harder pool, with success $q \ll p$. We derive the modulated exchange rate $R_q = R \cdot \frac{q(1-p)}{p(1-q)}$ and the flip threshold $q^* = \frac{(1-b)p}{(1-p)a+(1-b)p}$ below which *false rejections become the dominant harm.* Measured $q \approx 0.3$ against $q^* \approx 0.6$–0.75 predicts—from parameters alone (q itself estimated from a mixed rejection pool; §4)—that the LLM-judge arms in our study backfire; they do, most robustly the isolated self-judge (within-arm McNemar $p = 0.004$).

3. **Pooled-rate attribution (§3.6).** Pooled per-attempt operating points, the natural summary reported by most evaluation studies, underpredict terminal leak by a consistent $\approx 3.4\times$ across our judge arms (task-level bootstrap 95% CI $[2.1, 4.7]$). The cause is that pooled rates mask the per-round success trajectory p_k: a degradation-only model at the measured p_k reproduces the gap in full, leaving no residual that requires cross-instance miss clustering (the attribution split is convention-dependent and not separably identified at this sample size).

4. **Empirical validation and boundary conditions (§4).** Oracle verification confirms the theory exactly (against its own first attempts it never loses a correct solution and rescues 14/10 wrong ones at +27% cost); a matched-pool ablation provides suggestive (small-sample) evidence that actionable feedback raises repair success from 0.19 to 0.33; and a verifier-context experiment yields an honest negative result: sharing the generator's reasoning with a same-model judge did *not* collapse its accuracy, delimiting the scope of self-verification-collapse claims to self-correction loops rather than verification contexts.

2 Related Work

Self-correction and its critiques. Self-Refine (Madaan et al., 2023) and Reflexion (Shinn et al., 2023) popularized iterative self-improvement. Stechly et al. (2025) showed systematic collapse of self-critique on formal tasks, with LLM verifiers exhibiting very high false-rejection rates; Huang et al. (2024) and the critical survey of Kamoi et al. (2024) reached compatible conclusions. Our theory gives these observations a probabilistic form: collapse corresponds to failure-mode correlation

between generator and verifier (§3.3), and our experiment delimits where that correlation does and does not appear.

Test-time scaling theory. Dorner et al. (2026) prove that rejection sampling and Best-of-N accuracy is determined by verifier ROC geometry, and that RS dominates BoN at fixed compute. Their setting is pure resampling: no feedback, no repair, no generator–verifier correlation analysis, and no operating-point selection guidance. Our paper is a deliberate three-axis extension: repair feedback (§3.4), operating-point calibration (§3.2), and degraded/correlated retries (§3.5, 3.6). Related theory includes inference-time alignment coverage analysis (Huang et al., 2025). In the training-time regime, Plesner et al. (2026) study robustness of RLVR to reward noise and explicitly leave asymmetric (FN/FP) noise to future work; our Theorem 3 supplies the inference-time counterpart of that asymmetry in closed form.

Verification techniques. Self-consistency (Wang et al., 2023) uses agreement as the verification signal; process supervision (Lightman et al., 2024) trains step-level verifiers; LLM-as-judge evaluation is surveyed by Zheng et al. (2023). Control-theoretic formalizations of self-correction dynamics appear in Wu et al. (2026); they manage loop stability, whereas we manage the verifier's error budget. All of these techniques select a point on the verifier's quality trade-off; our results say *which* point to select and when the loop tolerates a harsh verifier.

3 Theory

3.1 Setup: the GVR loop

One *attempt*: a generator produces an artifact; a verifier either *accepts* or *rejects* it (with feedback); a rejection triggers a new attempt, up to budget N.

Definition 1 (Per-attempt rates)**.** *Let correctness of an attempt be binary, with generator success* $p = \Pr(C)$*. The verifier has acceptance rate* $a = \Pr(\textit{accept} \mid C)$ *and catch rate* $b = \Pr(\textit{reject} \mid \neg C)$*. Its two error types are the* miss $1 - b$ *(accepting wrong work) and the* false rejection $1 - a$ *(rejecting correct work).*

Definition 2 (Attempt outcomes)**.** *Per attempt:* $s = pa$ *(accept-correct),* $\ell = (1 - p)(1 - b)$ *(*leak*: accept-wrong), and* $r = 1 - s - \ell$ *(reject).*

Terminal states: *accepted* (split into correct and leak) and *unresolved* (budget exhausted). We assume (A1) the verdict is independent of the generation path given correctness and (A2) attempts are i.i.d. with constant (p, a, b); both are relaxed in §3.4–3.6.

Lemma 1 (Single-pass composition)**.** *For* $N = 1$*,* $\Pr(\textit{deliver wrong}) = (1 - p)(1 - b)$*: a wrong deliverable requires the conjunction of two independent failures. The false-rejection rate does not enter this quantity.*

Proof. Direct computation of $\Pr(\neg C \wedge \text{accept})$. □

Theorem 1 (Loop closed forms)**.** *Under (A1)–(A2), with budget* $N \geq 1$*:*

$$\Pr(\textit{leak}) = \ell \, \frac{1 - r^N}{1 - r}, \quad \Pr(\textit{accepted}) = (s + \ell)\frac{1 - r^N}{1 - r}, \tag{1}$$

$$\Pr(\textit{unresolved}) = r^N, \qquad \mathbb{E}[\textit{attempts}] = \frac{1 - r^N}{1 - r}. \tag{2}$$

Proof. The loop is geometric: reaching attempt k requires $k-1$ rejections (probability r^{k-1}); summing and telescoping gives the claims. The instance-level structure matches the ROC-based characterization of Dorner et al. (2026) for the pure resampling case; we state it in outcome-decomposed form for use below. □

Theorem 2 (Budget–precision invariance)**.** *For every $N \geq 1$,*

$$\pi := \Pr(\textit{wrong} \mid \textit{accepted}) = \frac{(1-p)(1-b)}{pa + (1-p)(1-b)},$$

independent of N: added budget converts unresolved mass into accepted mass at fixed precision.

Proof. Ratio of the leak and accepted probabilities in Theorem 1; the geometric factors cancel. □

3.2 The miss-asymmetry theorem

Both verifier errors hurt π—misses first-order through ℓ, false rejections second-order: each forced retry buys a fresh chance to leak. Their marginal damages have a closed-form ratio.

Theorem 3 (Miss-asymmetry)**.**

$$\frac{\partial\pi/\partial(1-b)}{\partial\pi/\partial(1-a)} = \frac{a}{1-b}.$$

Proof. $\partial\pi/\partial(1-b) = (1-p)s/(s+\ell)^2$ and $\partial\pi/\partial(1-a) = p\ell/(s+\ell)^2$ by the chain rule on Definition 2; the ratio simplifies to $a/(1-b)$. □

At $(a, b) = (0.95, 0.95)$, $R \approx 19$: one unit of catch rate is worth ∼19 units of acceptance rate for precision. Meanwhile the cost channels are reversed: $1-a$ enters expected attempts first-order, $1-b$ only through the negligible ℓ term.

Corollary 1 (Calibration guideline)**.** *Under (A2), any verifier modification trading acceptance for catch at an exchange rate better than* 1:R *strictly improves final precision; its price is paid only in attempts. Calibrate verifiers aggressively toward catch rate; budget false rejections rather than fear them. The qualifier is essential: when retries are degraded below q^* (Theorem 5) the effective exchange rate drops below 1 and the advice inverts—protect acceptance.*

Corollary 2 (Majority voting)**.** *Replace a verifier by a majority vote of k conditionally independent verifiers (strict majority rejects; ties accept). For $a, b > 1/2$ the vote raises* both *effective rates:*

$$b^{(k)} = \sum_{j>k/2} \binom{k}{j} b^j (1-b)^{k-j} > b, \qquad a^{(k)} = 1 - \sum_{j>k/2} \binom{k}{j} (1-a)^j a^{k-j} > a,$$

because majority rule makes false rejections rarer*, not commoner—a lone objecting judge no longer suffices to reject. Precision therefore improves through both channels, and generator attempts do not increase (the rejection mass r falls); the price is k verifier invocations per round: at the pooled self-judge operating point of §4, majority-of-three lowers expected generator attempts from* 1.38 *to* 1.25 *while raising expected verifier calls from* 1.38 *to* 3.75*—the cost moves from generation to verification, and whether it is worth paying depends on the two units' relative prices. The catch-vs-acceptance trade-off that R prices belongs instead to unanimity-style rules (reject if any judge objects), where $a^{(k)} = a^k$ falls sharply while $b^{(k)} = 1-(1-b)^k$ rises.*

3.3 Correlated failures

(A1) fails in a specific direction: a verifier that is the same model, primed with the generator's context, tends to repeat the generator's mistake. Model this with a latent failure-mode mixture.

Definition 3 (Failure-mode mixture)**.** *Conditioned on latent mode $M = m$ with prior π_m, generation succeeds with p_m; the verifier accepts correct output with a_m and catches wrong output with b_m, independently within mode.*

Definition 4 (Effective rates)**.** *With $\epsilon = \sum_m \pi_m(1 - p_m)$, error mass $\mu_m = \pi_m(1 - p_m)/\epsilon$ and success mass $\sigma_m = \pi_m p_m/(1 - \epsilon)$: $b_{\text{eff}} = \sum_m \mu_m b_m$ is the error-mass-weighted catch rate and $a_{\text{eff}} = \sum_m \sigma_m a_m$ the success-mass-weighted acceptance rate.*

Theorem 4 (Decorrelation condition)**.** *Attempt-1 outcomes obey the closed forms of §3.1 with pooled rates $(p, a, b) \to (1 - \epsilon,\, a_{\text{eff}},\, b_{\text{eff}})$; in particular the accepted-stream error*

$$\pi_1 = \frac{\epsilon\,(1 - b_{\text{eff}})}{\epsilon\,(1 - b_{\text{eff}}) + (1 - \epsilon)\,a_{\text{eff}}}$$

improves on no verification (delivered error ϵ) iff

$$a_{\text{eff}} + b_{\text{eff}} > 1,$$

i.e., iff the mass-weighted operating point lies above the chance diagonal of ROC space: verification helps exactly when $\Pr(\textit{accept} \mid C) > \Pr(\textit{accept} \mid \neg C)$. Catch alone does not suffice: if $b_{\text{eff}} \to 0$ while $a_{\text{eff}} < 1$, verification is strictly harmful *($\pi_1 = \epsilon/[\epsilon + (1 - \epsilon)a_{\text{eff}}] > \epsilon$), degenerating to neutral theater only in the rubber-stamp limit $a_{\text{eff}} \to 1$. Correlated failure has an exact form: with $\nu = \sum_m \pi_m(1 - b_m)$ the prior-weighted miss rate,*

$$\Pr(\textit{wrong} \wedge \textit{miss}) = \epsilon(1 - b_{\text{eff}}) = \epsilon\nu + \text{Cov}_m(1 - p_m,\, 1 - b_m),$$

the covariance over modes under prior π_m: decorrelation is precisely the claim that this covariance be small.

For $N > 1$ the mixture bites: with instance-persistent modes, surviving to round k conditions on the mode mix. §4 validates the attempt-1 form directly—every judge arm's pooled operating point lies above the chance diagonal, and every arm's single-shot delivered stream is purer than no verification—while the terminal gap beyond degraded retries is the quantity the covariance term would have to explain; our data bound it but do not identify it.

3.4 Repair feedback breaks i.i.d. — favorably

Proposition 1 (Repair factor)**.** *Let rejected attempts be followed by a repaired attempt succeeding with $p' = p + \delta(1 - p)$, $\delta \in [0, 1]$ ($\delta = 0$: rejection sampling; $\delta = 1$: perfect repair). Then all formulas of §3.2 with constant p are conservative (actual leak is upper-bounded), and the invariance of Theorem 2 breaks in the favorable direction: precision improves with budget.*

Proof sketch. Sample a uniform u once per attempt slot and couple a repaired draw (correct iff $u \le p'$) with a fresh one (correct iff $u \le p$); since $p' \ge p$, the repaired draw is correct whenever the fresh one is. Every round $k \ge 2$ therefore has masses $s' = p'a \ge s$ and $\ell' = (1 - p')(1 - b) \le \ell$, so its accepted stream is at least as pure, $\pi' = \ell'/(s' + \ell') \le \pi$, and every constant-$p$ formula is an upper bound on realized leak. The terminal conditional error at budget N is a weighted mean of π (round 1) and π' (rounds ≥ 2) with the π'-weight strictly increasing in N; it therefore decreases toward π'—the favorable break of Theorem 2's invariance. □

3.5 Repair degradation flips the asymmetry

Real repairs do not redraw at rate p: the retry pool is conditioned on rejection, dominated by hard instances, with success $q \ll p$.

Definition 5 (Retry rate)**.** $q := \Pr(\textit{attempt } k{+}1 \textit{ correct} \mid \textit{attempt } k \textit{ rejected}),\ k \geq 1.$

A falsely rejected *correct* solution enters a sub-loop with per-round success q; conditional on resolution it ends wrong with

$$\pi_q = \frac{(1-q)(1-b)}{qa + (1-q)(1-b)}.$$

Theorem 5 (Modulated exchange rate)**.** *At the population level, the marginal damage ratio becomes*

$$R_q \;=\; \frac{a}{1-b} \cdot \frac{q(1-p)}{p(1-q)}.$$

The asymmetry flips *(false rejections dominate) iff*

$$q < q^* \;=\; \frac{(1-b)p}{(1-p)a + (1-b)p}.$$

Proof. A unit ε of extra miss rate moves $\varepsilon(1-p)$ mass of wrong first attempts from "caught" (sub-loop wrong with probability π_q) to "delivered wrong" (probability 1): damage $\varepsilon(1-p)(1-\pi_q)$. A unit of extra false-rejection rate moves εp mass of correct first attempts from "delivered correct" to the sub-loop: damage $\varepsilon p \pi_q$. Substituting $\pi_q = \ell_q/(s_q + \ell_q)$ with $s_q = qa$, $\ell_q = (1-q)(1-b)$ and taking the ratio yields R_q; setting $R_q = 1$ and solving for q gives q^*. For $q = p$ the expression recovers Theorem 3. □

Corollary 3 (Backfire prediction)**.** *If a verifier has $1-a$ non-negligible and measured $q < q^*$, adding it to the loop reduces terminal accuracy below the no-verification baseline, despite any plausible catch rate. The conversion rate of correct solutions into wrong ones is $(1-a)\,p\,\pi_q$.*

3.6 Clustering and pooled-rate failure

The pooled $(\hat{a}, \hat{b})$ reported by evaluation studies summarize attempts across instances. Two measured phenomena make pooled rates poor predictors of terminal leak: (i) retry degradation (§3.5) multiplies per-retry leak chance from $(1-p)(1-b)$ to $(1-q)(1-b)$; (ii) misses cluster on particular wrong-solution instances that recur across rounds, so instance-level modes with low b_m carry disproportionate mass. Both effects push in the same direction; §4 quantifies a consistent $\approx 3.4\times$ underprediction of leak by pooled-rate theory and shows that (i) alone reproduces it, leaving (ii) unidentified at this sample size.

4 Experiments

Setup. HumanEval (Chen et al., 2021), all 164 problems. Generator: qwen-plus (temperature 1.0, max 1024 tokens), prompted to plan then emit a Python block. Budget $N = 5$. Six arms: none (deliver first attempt); oracle_full (unit tests run in a subprocess sandbox; feedback = failing-test traceback); oracle_verdict (tests; feedback = "failed" only); judge_hetero (qwen-max judge, isolated context); judge_self_iso (qwen-plus judge, isolated context); judge_self_ctx (qwen-plus judge shown the generator's own reasoning). Every attempt is additionally scored by the oracle *silently* for instrumentation, so per-arm operating points $(\hat{a}, \hat{b})$, per-round success p_k, and leak/terminal outcomes are all measured directly. All 984 runs completed; total cost 1.64M tokens.

Table 1: Main results (n=164 per arm; 95% Wilson CIs). π_{pred} from Theorem 2 with measured $\hat{p}_1, \hat{a}, \hat{b}$; π_{obs} = error conditional on delivery.

Arm	$\hat{p}_1$	$\hat{a}$	$\hat{b}$	Acc.	Leak	Unres.	$\mathbb{E}[k]$	π_{pred}	π_{obs}
none	.91	–	–	.91	.091	.000	1.00	–	–
oracle_full	.89	1.00	1.00	.98	.000	.024	1.27	.000	.000
oracle_verdict	.90	1.00	1.00	.96	.000	.037	1.29	.000	.000
judge_hetero	.90	.78	.77	.85	.098	.049	1.52	.030	.103
judge_self_iso	.90	.78	.74	.82	.116	.061	1.51	.037	.123
judge_self_ctx	.90	.86	.86	.87	.061	.067	1.45	.018	.065

Table 2: Within-arm pairing: terminal outcome vs. the same arm's first attempt, by task (McNemar exact). "Lost" = first attempt correct, terminal not delivered-correct; "rescued" = first attempt wrong, terminal correct. p: unresolved counted as failure (the paper's accuracy convention); p_{del}: unresolved tasks excluded. Effect = terminal accuracy minus own first-attempt accuracy.

Arm	Lost	Rescued	Net	p	p_{del}	Effect
oracle_full	0	14	+14	.0001	.0001	+8.5pp
oracle_verdict	0	10	+10	.002	.002	+6.1pp
judge_hetero	10	2	−8	.039	.29	−4.9pp
judge_self_iso	14	2	−12	.004	.013	−7.3pp
judge_self_ctx	10	6	−4	.46	1.00	−2.4pp

4.1 Oracle verification: theory confirmed exactly

Unit-test verification lifts accuracy from .91 to .98 at +27% attempts ($\mathbb{E}[k] = 1.27$), with leak collapsing from 9.1% to exactly 0.0%, as Theorem 1 requires for $b = 1$. Within its own first attempts the loop never loses a correct solution and rescues 14/10 wrong ones (Table 2). This is the clean empirical instantiation of Corollary 1: at $a = 1$ its q-qualification is vacuous (there are no false rejections to convert), and budget spent on catch rate pays off at near-parity cost.

4.2 LLM judges lose correct solutions, as the degradation theorem predicts

All three judge arms land *below* the no-verification baseline (.85/.82/.87 vs. .91) despite pooled operating points that look respectable. Cross-arm comparisons, however, are confounded by first-draw sampling: each arm drew its first attempts independently, and $\hat{p}_1$ ranges .89–.91 (none happened to draw the best of the six). The causal comparison pairs each arm's terminal outcome against *its own* first attempt on the same task.

Table 2 is the paper's cleanest contrast. Oracle verification never loses a correct solution (a unit acceptance rate never re-rolls correct work) and rescues 14/10 wrong first attempts. The isolated self-judge does the reverse, significantly under both conventions ($p = .004$ and .013)—the one backfire robust to how unresolved tasks are counted. The cross-capability judge is significant only under the accuracy convention (it fails a Bonferroni correction across the three judge arms and loses significance when unresolved tasks are excluded); the context-sharing judge's deficit is a null. We accordingly claim a significant backfire only for judge_self_iso.

The mechanism is measured directly: false rejections send correct solutions into a retry sub-loop whose conditional success is $q \approx 0.27$–0.39 (Table 3), so correct→wrong conversion occurs at the rate Corollary 3 predicts, $(1 - \hat{a})\hat{p}_1\hat{\pi}_q$ with $q = \hat{p}_2$: predicted 15.7/5.6/10.0 conversions

Table 3: Per-round conditional success p_k (rows: arms; $n \geq 10$). $q := \hat{p}_2$ (second-round conditional success), the quantity used in §4; judge_self_ctx's $p_5 = .00$ rests on 11 rounds in a heavily depleted pool.

Arm	p_1	p_2	p_3	p_4	p_5
oracle_full	.89 (n=164)	.33 (18)	.25 (12)	–	–
oracle_verdict	.90 (164)	.19 (16)	.23 (13)	.10 (10)	–
judge_hetero	.90 (164)	.39 (36)	.42 (24)	.13 (15)	.45 (11)
judge_self_iso	.90 (164)	.27 (37)	.37 (19)	.35 (17)	.40 (10)
judge_self_ctx	.90 (164)	.30 (30)	.32 (19)	.29 (14)	.00 (11)

Table 4: Conditional-on-delivery error π by budget N (runs truncated at N rounds). The no-verification baseline (deliver the first attempt unverified) has $\pi = .091$.

Arm	N=1	N=2	N=3	N=4	N=5
judge_hetero	.031	.050	.074	.092	.103
judge_self_iso	.024	.103	.109	.123	.123
judge_self_ctx	.022	.055	.067	.065	.065

for self_iso/self_ctx/hetero vs. observed 12/6/6—the correct order and the correct arm ranking. Predictions are expectations; with ∼30 expected events the binomial noise is ±5. (Measured $\hat{p}_2$ pools rejections of genuinely wrong and of correct attempts—55 vs. 39 rejection events in hetero—so it approximates, rather than identifies, the correct-solution sub-loop rate; cf. Limitations, §5.) The flip thresholds are $q^* \approx 0.75/0.59/0.73$; measured q lies far below all of them, so the backfire is predicted from parameters alone, before any end-to-end run.

4.3 Single-shot verification helps; retries eat the gain

Truncating every run at budget N and recomputing conditional-on-delivery error traces the loop's purity curve (Table 4).

Two readings. First, at $N = 1$ every judge arm's delivered stream is far purer than the no-verification baseline (.022–.031 vs. .091), exactly as Theorem 4 requires: each arm's pooled operating point lies above the chance diagonal ($\hat{a} + \hat{b} = 1.52$–1.72), so single-shot verification helps. Second, purity degrades with budget: retries draw at $q \ll p$, so each additional round admits increasingly impure accepted mass. Budget is thus *not neutral*: Theorem 2's invariance holds only when retries redraw at the generator's rate—Proposition 1 shows it breaking favorably when repairs improve, the degradation of §3.5 unfavorably when they do not—and the sweep realizes the unfavorable side, where added budget *actively* worsens the delivered stream. For judge_self_iso the first retry alone erases the entire single-shot gain (π: .024 → .103, crossing the baseline .091). Budget, like strictness, is a q-dependent decision (G2, §5).

4.4 Pooled operating points underpredict leak by ≈ 3.4×—fully explained by degraded retries

Using measured pooled $(\hat{a}, \hat{b})$ and $\hat{p}_1$, Theorem 2 predicts conditional-on-delivery error of 3.0%/3.7%/1.8%; observed are 10.3%/12.3%/6.5%—per-arm ratios of 3.4/3.3/3.6, geometric mean 3.4 (task-level bootstrap 95% CI $[2.1, 4.7]$, far from 1) (Table 1, last two columns). What does this gap mean? A degradation-only model—the measured per-round trajectory p_k (Table 3) with constant $(\hat{a}, \hat{b})$,

no instance-level clustering—reproduces it in full: predicted π of .106/.134/.068 vs. observed .103/.123/.065, slightly *over*predicting every arm. Degraded retries are thus not merely the dominant channel but a *sufficient* one: the data require no clustering channel. The precise attribution share, however, is a modeling convention rather than a measurement: across the two conventions we implement (retries at constant $q = \hat{p}_2$ vs. at the measured p_k; per-round survival) it spans 98–128%, and at this sample size degradation and clustering are not separably identified; a per-instance binomial overdispersion check ($\chi^2/\mathrm{df} = 1.3$–2.2 across judge arms; the same wrong instances do leak at k=1 through k=5) is directionally consistent with mild miss clustering but does not separate the channels. The leak composition confirms that degradation carries the gap: of the 16 leaks in judge_hetero, 6 arise from falsely rejected *correct* first attempts—the conversion channel of Theorem 5—and 10 from wrong first attempts that survived repeated verification; in the two self-judge arms the conversion channel is the majority (12/19 and 6/10). Instance-level modeling, as in Dorner et al. (2026), remains a necessity for terminal-error prediction; but the first-order lesson is simpler—pooled rates fail because they mask the per-round trajectory, so measure q.

4.5 Actionable feedback repairs better (matched pools)

oracle_full vs. oracle_verdict share the same retry pool (first-attempt-wrong problems), isolating feedback content; with $b = 1$ the pool is exactly the wrong-first-attempt tasks, so this comparison does not inherit the mixed-pool approximation of the judge arms. Traceback feedback raises second-round repair success from .19 to .33 (+74% relative; 3/16 vs. 6/18 events, Fisher exact $p = 0.45$) and lowers unresolved rate (.024 vs. .037). The sample is small, so we read this as *suggestive* evidence on a literature tension rather than an adjudication: Stechly et al. (2025) found feedback detail immaterial with sound verifiers on formal tasks, while Reflexion-style systems premise feedback-driven repair; our δ-framework (Proposition 1) can accommodate both regimes, and code repair appears to sit in the $\delta > 0$ regime.

4.6 An honest negative result on context-induced collapse

Contrary to the collapse motivation of §3.3, the judge *shown the generator's own reasoning* (judge_self_ctx) was slightly *more* accurate at both operating points than the isolated same-model judge ($\hat{a} = .86$ vs. .78; $\hat{b} = .86$ vs. .74) and delivered better terminal accuracy (.87 vs. .82). These gaps are non-significant—the $\hat{b}$ Wilson intervals overlap ($[.76, .92]$ vs. $[.63, .83]$)—and the context arm's within-arm deficit is the weakest of the three (net -4, $p = 0.46$; $p = 1.0$ counting only delivered outcomes), so we claim no evidence of context-induced collapse but cannot exclude a moderate harmful effect either. In this setting, context-sharing did not corrupt verification; collapse phenomena appear confined to self-*correction* loops (Stechly et al., 2025), not verifier contexts. We report this as a boundary condition on the theory: the correlation that destroys verification loops must be measured (Theorem 4), not assumed.

5 Discussion

Design guidelines. G1: oracle-first. Wherever a deterministic check exists (tests, compilers, schemas), attach it to the verifier: it removes leak entirely at bounded cost. **G2: strictness must be earned by repairability.** Harsh verification is safe when retries are cheap and successful (q high); when repairs are degraded, Corollary 1 inverts—acceptance must be protected. Measure q (one rejected-batch pilot) before choosing a verifier's operating point *or its budget* (§4.3). **G3:**

never trust pooled operating points for terminal-error prediction; report instance-level effects. **G4: engineer feedback actionability (δ)**—it is the cheapest lever on repair success.

Human analogy. The same structure appears in human peer review: independent review layers compose multiplicatively (the "Swiss cheese" model of accident analysis), reviewer strictness is affordable exactly when authors can repair well, and nitpicking a correct solution has its price paid in rework—sometimes turning a working change into a broken one, our conversion mechanism in miniature.

Limitations. Single generator family (Qwen) and one benchmark (HumanEval, 164 problems); judge "heterogeneity" is cross-capability, not cross-family; leak rates rest on 10–19 instances per judge arm. The feedback ablation rests on 3 vs. 6 repair events (Fisher $p = 0.45$) and is suggestive only (its retry pool, unlike the judge arms', is cleanly identified); the retry rate q for the judge arms is measured from a mixed rejection pool and only approximates the correct-solution sub-loop rate; the context-collapse negative result is one setting with overlapping confidence intervals. The theory's identifiability assumptions (within-mode independence) are idealizations. The artifact code supports replication and extension.

6 Conclusion

Verification is not free reliability. Its benefit is governed by a computable exchange rate between the verifier's two error types; its danger—demonstrated here empirically at the rates the theory predicts—appears when degraded repairs let false rejections convert correct work into wrong. The practical summary fits in one sentence: *spend your verifier budget on catching errors, but first measure how well your generator recovers from being told it was wrong.*

Reproducibility. Experiment code, prompts, per-attempt logs (984 runs, 1.64M tokens), and analysis scripts are released at `https://github.com/dianxia/verifier-backfire`.

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